

CSA0804 PYTHON CO2 AT3 CODING CHALLENGE

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Student Submissions — CO2_AT3_Coding Challenge

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Q1. String Character Frequency Counter

CODE

```
s = input().lower()
d = {}
for ch in s:
    if ch != " ":
        if ch in d:
            d[ch] += 1
        else:
            d[ch] = 1
for ch in sorted(d):
    print(ch, "→", d[ch])
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()
vowels = "aeiou"
vc=cc=0
vf={}
cf={}
for ch in s:
    if ch.isalpha():
        if ch in vowels:
            vc+=1
            vf[ch]=vf.get(ch,0)+1
        else:
            cc+=1
            cf[ch]=cf.get(ch,0)+1
print("Vowels:",vc)
print("Consonants:",cc)
if vf:
    mv=max(vf, key=vf.get)
    print("Most frequent vowels",mv,"(",vf[mv],")",sep="")
else:
    print("Most frequent vowel:None")
if cf:
    lc=min(cf, key=cf.get)
    print("Least frequent consonant",lc,"(",cf[lc],")",sep="")
else:
    print("Least frequent consonant:None")
if cc==0:
    print("Vowel-to-consonant Ratio:0.00")
else:
```

```
print("Vowel-to-consonant Ratio:0.00")
else:
    print("Vowel-to-consonant Ratio:",format(vc/cc,"zf"))
```

INPUT Python Programming

OUTPUT

```
Vowels: 4
Consonants: 13
Most frequent vowelso(2)
Least frequent consonanty(1)
Vowel-to-consonant Ratio: 0.307692
```

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Q3. String Compression and Run-Length Encoding

CODE

```
s=input()
c=""
count=1
for i in range(len(s)):
    if i<len(s)-1 and s[i]==s[i+1]:
        count+=1
    else:
        c+=s[i]
        if count>1:
```

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s=input()
c=""
count=1
for i in range(len(s)):
    if i<len(s)-1 and s[i]==s[i+1]:
        count+=1
    else:
        c+=s[i]
        if count>1:
            c+=str(count)
        count=1
if len(c)<len(s):
    print("Compressed:",c)
else:
    print("Compression not beneficial")
    print("Original",s)
d=""
i=0
while i<len(c):
    ch=c[i]
    i+=1
    num=""
    while i<len(c) and c[i].isdigit():
        num+=c[i]
        i+=1
    if num=="":
```

```

while i<len(c):
    ch=c[i]
    i+=1
    num=""
    while i<len(c) and c[i].isdigit():
        num+=c[i]
        i+=1
    if num!="":
        d+=ch
    else:
        d+=ch*int(num)
print("Decompressed:",d)

```

INPUT

aabbccdddee

OUTPUT

Compressed: a2b3c2d4e2
Decompressed: aabbccdddee

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Q4. Advanced Text Statistics Analyser

CODE

```

text = input()
stop = ("the", "is", "a", "an", "in", "of", "and", "to", "for")
words = text.split()
sent = text.count(".") + text.count("!") + text.count("?")
if sent == 0:
    sent = 1
print("Words:", len(words))
print("Sentences:", sent)
print("Paragraph: ", 1)
total = 0
for w in words:
    total += len(w.strip(".!?"))
print("Avg word Length:", round(total/len(words),2))
print("Avg Sentence Length:", round(len(words)/sent,2))
freq = {}
for w in words:
    w = w.lower().strip(".!?")
    if w not in stop:
        if w in freq:
            freq[w] += 1
        else:
            freq[w] = 1
print("Top 5 words:")
for i in sorted(freq, key=freq.get, reverse=True) [:5]:
    print(i, "(", freq[i],")", sep="")
vowels = 0

```

```

print("Top 5 words:")
for i in sorted(freq, key=freq.get, reverse=True) [:5]:
    print(i, "(", freq[i],")", sep="")
vowels = 0
for w in words:
    for ch in w.lower():
        if ch in "aeiou":
            vowels += 1
score = 206.835 - 1.015*(len(words)/sent) - 84.6*(vowels/len(words))
print("Flesch Reading Ease:", round(score,1))

```

INPUT

Python is a powerful programming language

OUTPUT

Words: 6
Sentences: 1
Paragraph: 1
Avg word Length: 6.0
Avg Sentence Length: 6.0
Top 5 words:
python(1)
powerful(1)
programming(1)
language(1)
Flesch Reading Ease: 17.4